Generosity vs. Access: The Effects of Subsidy Policy on Price and Quality of Child Care

Serena Goldberg

November 3, 2026

Motivation

- Quality care improves low-income children's long-run outcomes (e.g., Almås, Drange, Meghir, and Zachrisson 2026; Bailey, Sun and Timpe 2021; Deming 2009)

Motivation

- Quality care improves low-income children's long-run outcomes (e.g., Almås, Drange, Meghir, and Zachrisson 2026; Bailey, Sun and Timpe 2021; Deming 2009)
- Child Care and Development Fund aims to enroll more low-income children in high-quality care (Largest child care program in US: $12.4 billion/yr)
  CCDF Statute

Motivation

- Quality care improves low-income children's long-run outcomes (e.g., Almås, Drange, Meghir, and Zachrisson 2026; Bailey, Sun and Timpe 2021; Deming 2009)
- Child Care and Development Fund aims to enroll more low-income children in high-quality care (Largest child care program in US: $12.4 billion/yr)
  CCDF Statute
  - Funds state subsidy programs

Motivation

- Quality care improves low-income children's long-run outcomes (e.g., Almås, Drange, Meghir, and Zachrisson 2026; Bailey, Sun and Timpe 2021; Deming 2009)
- Child Care and Development Fund aims to enroll more low-income children in high-quality care (Largest child care program in US: $12.4 billion/yr)
  CCDF Statute
  - Funds state subsidy programs
  - States must trade off: serve more families? Or offer larger subsidies?

This Paper

- With a larger subsidy, subsidized families can afford high-quality care

This Paper

- With a larger subsidy, subsidized families can afford high-quality care
- But if funding is limited, what happens to families who lose out on the subsidy?

This Paper

- With a larger subsidy, subsidized families can afford high-quality care
- But if funding is limited, what happens to families who lose out on the subsidy?
- As demand shifts, market prices, quality change in equilibrium

This Paper

- With a larger subsidy, subsidized families can afford high-quality care
- But if funding is limited, what happens to families who lose out on the subsidy?
- As demand shifts, market prices, quality change in equilibrium
- This paper: What should be the structure of the subsidy program?

What I Do

- Novel administrative data from Wisconsin
  - Near universe of licensed private care, Head Start, public 4-K

What I Do

- Novel administrative data from Wisconsin
- First: quasi-experimental evidence
  - Identify effect of how much the state offers to subsidized families and how many families are served on local price and quality
  - Leverage variation over space, time, provider type

What I Do

- Novel administrative data from Wisconsin
- First: quasi-experimental evidence
  - Identify effect of how much the state offers to subsidized families and how many families are served on local price and quality
  - Leverage variation over space, time, provider type
  - Neighborhoods with more subsidized families are more likely to have high-quality providers (1.3pp, 6.3% of sample mean)
  - Higher maximum subsidy leads to higher prices (5% higher for every $10)

What I Do

- Novel administrative data from Wisconsin
- First: quasi-experimental evidence
- But insufficient to answer our original question
    - Are low-income families better off when the state serves more families or offers a larger maximum subsidy?

What I Do

- Novel administrative data from Wisconsin
- First: quasi-experimental evidence
- But insufficient to answer our original question
  - Are low-income families better off when the state serves more families or offers a larger maximum subsidy?
  - How many fewer families are subsidized if the state increases the maximum subsidy?

What I Do

- Novel administrative data from Wisconsin
- First: quasi-experimental evidence
- But insufficient to answer our original question
  - Are low-income families better off when the state serves more families or offers a larger maximum subsidy?
  - How many fewer families are subsidized if the state increases the maximum subsidy?
  - How much do these changes matter?

What I Do

- Novel administrative data from Wisconsin
- First: quasi-experimental evidence
- But insufficient to answer our original question
- Second: Model with endogenous prices, quality, number of subsidies
    - Estimate to match enrollment, quality distribution, other moments

What I Do

- Novel administrative data from Wisconsin
- First: quasi-experimental evidence
- But insufficient to answer our original question
- Second: Model with endogenous prices, quality, number of subsidies
  - Estimate to match enrollment, quality distribution, other moments
  - Relative to subsidizing more families, a higher max subsidy pushes families from "middling-quality" private care to no formal care
  - Lower consumer, producer surplus (17%, 2%)

Literature

- Child care subsidy policy in equilibrium (Berlinski et al. 2024; Borowsky et al. 2022; Bodéré 2023; **Primary school:** Neilson 2025)
  - Novel data on near-universe of licensed and public care
  - Account for variation in subsidized enrollment over space
  - Cost-equivalent policies where not all eligible families can be subsidized

Literature

- Child care subsidy policy in equilibrium
- Improving the quality of child care for low-income children (Currie & Neidell 2007; Walters 2015; Francis & Barnett 2019; Joo et al. 2020; Drange & Telle 2018; Herbst 2018; **Subsidies:** Herbst & Tekin 2016, Krafft, Davis, and Trout 2017)
  - Subsidy policy design can "crowd in" quality

Literature

- Child care subsidy policy in equilibrium
- Improving the quality of child care for low-income children
- Social safety net program design (**Child care:** Lee, Sojourner, Davis & Borowsky 2025; Ho & Pavoni 2020; **Health:** Cabral, Geruso & Mahoney 2018; Duggan, Starc & Vabson 2016; Finkelstein 2007; **Housing:** Collinson & Ganong 2018; Eriksen & Ross 2015; Fack 2006; Susin 2002; **SNAP:** Finkelstein & Notowidigdo 2019; Leung & Seo 2023; **EITC:** Rothstein 2010; **Program Targeting:** Rafkin, Solomon & Soltas 2023; **Optimal Tax & Transfer:** e.g., Blundell & Shephard 2012; Saez 2002)
  - Evaluate how local participation affects price and quality

Introduction

Context and Data

Quasi-Experimental Evidence

Model

Estimation and Counterfactuals

Conclusion

Introduction

Context and Data
- Wisconsin's Child Care Market
- Subsidy Use is Concentrated Across Space
- WI Policy 2017-2025
- Data

Quasi-Experimental Evidence

Model

Estimation and Counterfactuals

Conclusion

The Child Care Market in Wisconsin

- 64% of enrollees attend private care
    - Majority are centers: operate from commercial facilities, high capacity

The Child Care Market in Wisconsin

- 64% of enrollees attend private care
  - Majority are centers: operate from commercial facilities, high capacity
  - Alternative are family providers: operate from residential facilities, low capacity

The Child Care Market in Wisconsin

- 64% of enrollees attend private care
- WI has robust public care
  - 4-K
    - Provided by most school districts
    - Generally only available for four-year-olds
    - Limited schedules
    - Optional: 63% of four-year-olds enrolled

The Child Care Market in Wisconsin

- 64% of enrollees attend private care
- WI has robust public care
  - 4-K
  - Head Start
    - Only children below the poverty line are eligible
    - Generally serve children ages 3-4

Introduction

Context and Data
- Wisconsin's Child Care Market
- Subsidy Use is Concentrated Across Space
- WI Policy 2017-2025
- Data

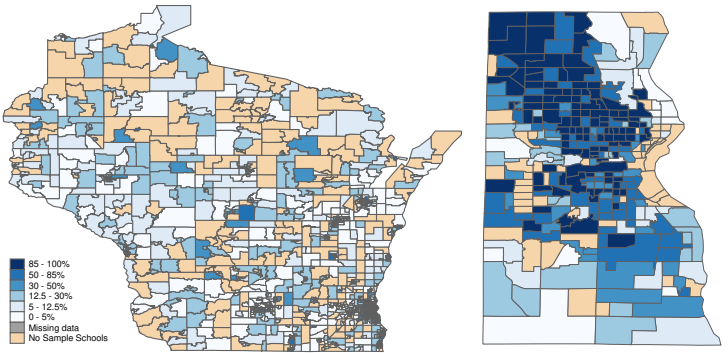
Quasi-Experimental Evidence

Model

Estimation and Counterfactuals

Conclusion

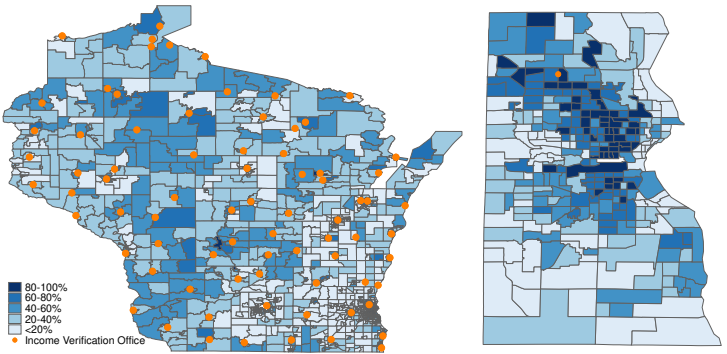
Subsidy Use is Concentrated Across Space



(a) Wisconsin (b) Milwaukee

Figure: Proportion of Children Enrolled in Private Care with Subsidies by Census Tract

Partially, Because Poverty Is Concentrated Across Space

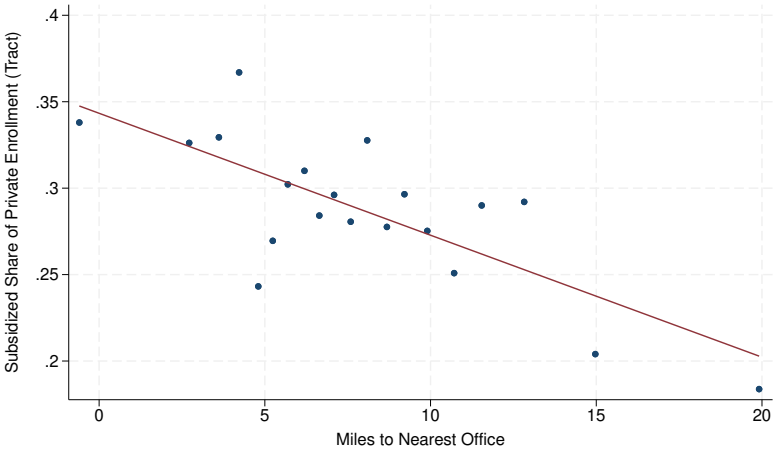


(a) Wisconsin

(b) Milwaukee

Figure: Proportion of Families Eligible for Subsidies (2025)

Subsidy Use Also Driven by Distance to Verification Offices



Outcome is the proportion of enrollment in the census tract from subsidized families (2022-2025). Controls for county, urbanicity, and the proportion of families eligible for subsidies.

Introduction

Context and Data
- Wisconsin's Child Care Market
- Subsidy Use is Concentrated Across Space
- WI Policy 2017-2025
- Data

Quasi-Experimental Evidence

Model

Estimation and Counterfactuals

Conclusion

How do Subsidies Work in WI?

- Families can choose between any licensed, quality-rated provider

How do Subsidies Work in WI?

- Families can choose between any licensed, quality-rated provider
- State pays price of care up to a maximum subsidy Detail

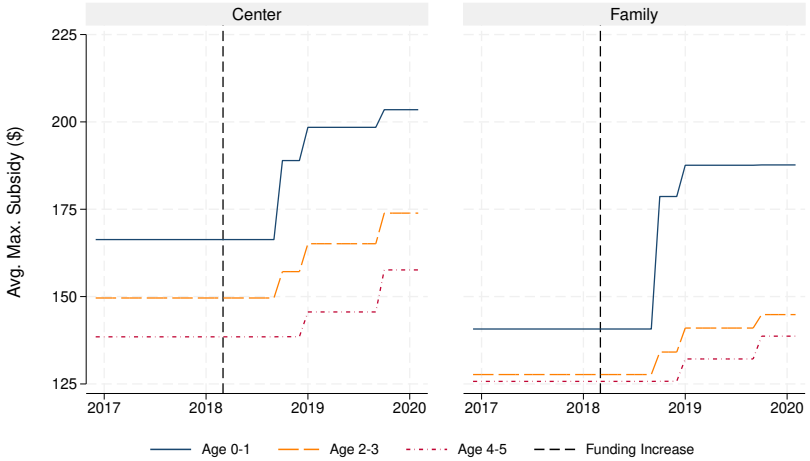
How do Subsidies Work in WI?

- Families can choose between any licensed, quality-rated provider
- State pays price of care up to a maximum subsidy Detail
- Max set by county, child age, facility provider operates from

How do Subsidies Work in WI?

- Families can choose between any licensed, quality-rated provider
- State pays price of care up to a maximum subsidy Detail
- Max set by county, child age, facility provider operates from
- Federal funding is set by formula and Congressional allocations

Weekly Maximum Subsidy Increased Sharply in 2018-2019



WI Increases Maximum Subsidy Again in 2022

- Congress again increased CCDF funding by 15% (FY2019-2021)
- WI increased the maximum subsidy in 2022:
  - Average increase across the state of $\sim$ 130%
  - New average subsidy of $\sim$1.6k/month (per family, at least half of family income for 79% of participating families)

Introduction

Context and Data
- Wisconsin's Child Care Market
- Subsidy Use is Concentrated Across Space
- WI Policy 2017-2025
- Data

Quasi-Experimental Evidence

Model

Estimation and Counterfactuals

Conclusion

Admin Data for Quasi-Experimental Evidence

- Need a clean pre-period: leverage 2018-2019 max subsidy change

Admin Data for Quasi-Experimental Evidence

- Need a clean pre-period: leverage 2018-2019 max subsidy change
- Merge admin data
  - Prices Detail
  - Quality Detail
  - Subsidized enrollment Detail
  - Addresses and other characteristics Detail

Admin Data for Quasi-Experimental Evidence

- Need a clean pre-period: leverage 2018-2019 max subsidy change
- Merge admin data
- Private, licensed, year-round, subsidy-accepting providers

Admin Data for Quasi-Experimental Evidence

- Need a clean pre-period: leverage 2018-2019 max subsidy change
- Merge admin data
- Private, licensed, year-round, subsidy-accepting providers
- Use American Community Survey for tract-level characteristics, demographics

Timeline

Additional Data Available After 2020

- Can merge enrollment data available starting in 2020 Detail

Additional Data Available After 2020

- Can merge enrollment data available starting in 2020 Detail
- Use data from 2022-2025 to estimate model

Additional Data Available After 2020

- Can merge enrollment data available starting in 2020 Detail
- Use data from 2022-2025 to estimate model
- Add private providers who do not accept subsidies, public providers (4-K/Head Start) Detail
- Bring in additional NSECE survey data to discipline the estimates Detail

Sample Types | Private Providers | Public 4-K Detail | Sample Comparison | Timeline

Introduction

Context and Data

Quasi-Experimental Evidence

Model

Estimation and Counterfactuals

Conclusion

Overview and Preview of Findings

- Do prices, quality respond to changes in the proportion of families subsidized or the maximum subsidy?

Overview and Preview of Findings

- Do prices, quality respond to changes in the proportion of families subsidized or the maximum subsidy?
- I use two sources of variation to disentangle:
  - Distance to income verification offices
  - Variation in maximum subsidy over time, by county, provider type

Overview and Preview of Findings

- Do prices, quality respond to changes in the proportion of families subsidized or the maximum subsidy?
- I use two sources of variation to disentangle:
  - Distance to income verification offices
  - Variation in maximum subsidy over time, by county, provider type
- Tracts with more subsidized families are more likely to have a high-quality provider

Overview and Preview of Findings

- Do prices, quality respond to changes in the proportion of families subsidized or the maximum subsidy?
- I use two sources of variation to disentangle:
  - Distance to income verification offices
  - Variation in maximum subsidy over time, by county, provider type
- Tracts with more subsidized families are more likely to have a high-quality provider
- Higher maximum subsidy leads to higher out-of-pocket payments

Introduction

Context and Data

Quasi-Experimental Evidence
- Quasi-Experimental Methods
- Quality
- Out-of-Pocket Payments

Model

Estimation and Counterfactuals

Conclusion

Why 2SLS-DiD

- Proportion of families receiving subsidies reflects unobserved demand for (quality) care

Why 2SLS-DiD

- Proportion of families receiving subsidies reflects unobserved demand for (quality) care
  - IV: distance to nearest income verification office
  - Exogenous conditional on local SNAP participation, demographics, urbanicity LATE
  - Maximum subsidy may affect demand for the subsidy, so need to control for both

Why 2SLS-DiD

- Proportion of families receiving subsidies reflects unobserved demand for (quality) care
- Maximum subsidy is higher in counties with higher prices

Why 2SLS-DiD

- Proportion of families receiving subsidies reflects unobserved demand for (quality) care
- Maximum subsidy is higher in counties with higher prices
  - Leverage 2018-2019 roll-out across time, county, provider type
  - Treat as exogenous conditional on county, year, provider-type FE

ATE on MaxSub

2SLS-DiD First Stage

- IV for proportion of families receiving subsidy (and interaction Interaction)

2SLS-DiD First Stage

- IV for proportion of families receiving subsidy (and interaction Interaction)

$$
\begin{aligned}
PrSub_{lt} = & \alpha_0 + \alpha_1 DistOffice_i + \alpha_2 MaxSub_{it} \\
& + \alpha_3 (DistOffice_i \times MaxSub_{it}) + \alpha_4 X_{it} \qquad (1a) \\
& + \zeta_c + \zeta_t + \nu_{it}
\end{aligned}
$$

2SLS-DiD First Stage

- IV for proportion of families receiving subsidy (and interaction Interaction)

$$\begin{aligned} PrSub_{lt} =& \alpha_0 + \alpha_1 DistOffice_i + \alpha_2 MaxSub_{it} \\ &+ \alpha_3 (DistOffice_i \times MaxSub_{it}) + \alpha_4 X_{it} \\ &+ \zeta_c + \zeta_t + \nu_{it} \end{aligned} \tag{1a}$$

- $l(i)$: census tract

2SLS-DiD First Stage

- IV for proportion of families receiving subsidy (and interaction Interaction)

$$\begin{aligned} PrSub_{lt} =& \alpha_0 + \alpha_1 DistOffice_i + \alpha_2 MaxSub_{it} \\ &+ \alpha_3 (DistOffice_i \times MaxSub_{it}) + \alpha_4 X_{it} \\ &+ \zeta_c + \zeta_t + \nu_{it} \end{aligned} \tag{1a}$$

- t: year

2SLS-DiD First Stage

- IV for proportion of families receiving subsidy (and interaction Interaction)

$$\begin{aligned} PrSub_{lt} =& \alpha_0 + \alpha_1 DistOffice_i + \alpha_2 MaxSub_{it} \\ &+ \alpha_3 (DistOffice_i \times MaxSub_{it}) + \alpha_4 X_{it} \qquad (1a) \\ &+ \zeta_c + \zeta_t + \nu_{it} \end{aligned}$$

- i: unit of observation
  - Census tract (any quality provider), weighted by number of young children
  - Provider (out-of-pocket payment), weighted by sub. enrollment

2SLS-DiD First Stage

- IV for proportion of families receiving subsidy (and interaction Interaction)

$$\begin{aligned} PrSub_{lt} =& \alpha_0 + \alpha_1 DistOffice_i + \alpha_2 MaxSub_{it} \\ &+ \alpha_3 (DistOffice_i \times MaxSub_{it}) + \alpha_4 X_{it} \\ &+ \zeta_c + \zeta_t + \nu_{it} \end{aligned} \qquad (1a)$$

- $PrSub_{lt}$: number of subsidized children using care in tract l divided by number of children under five in the tract, year t

2SLS-DiD First Stage

- IV for proportion of families receiving subsidy (and interaction Interaction)

$$\begin{aligned} PrSub_{lt} =& \alpha_0 + \alpha_1 DistOffice_i + \alpha_2 MaxSub_{it} \\ &+ \alpha_3 (DistOffice_i \times MaxSub_{it}) + \alpha_4 X_{it} \\ &+ \zeta_c + \zeta_t + \nu_{it} \end{aligned} \tag{1a}$$

- $DistOffice_i$: distance to nearest income-verification office

2SLS-DiD First Stage

- IV for proportion of families receiving subsidy (and interaction Interaction)

$$\begin{aligned} PrSub_{lt} =& \alpha_0 + \alpha_1 DistOffice_i + \alpha_2 MaxSub_{it} \\ &+ \alpha_3 (DistOffice_i \times MaxSub_{it}) + \alpha_4 X_{it} \\ &+ \zeta_c + \zeta_t + \nu_{it} \end{aligned} \tag{1a}$$

- ζ_c, ζ_t: county and year fixed effects

2SLS-DiD First Stage

- IV for proportion of families receiving subsidy (and interaction Interaction)

$$\begin{aligned} PrSub_{lt} =& \alpha_0 + \alpha_1 DistOffice_i + \alpha_2 MaxSub_{it} \\ &+ \alpha_3 (DistOffice_i \times MaxSub_{it}) + \alpha_4 X_{it} \\ &+ \zeta_c + \zeta_t + \nu_{it} \end{aligned} \tag{1a}$$

- X_{it}: provider type, local SNAP, demographics

2SLS-DiD Second Stage

$$
\begin{aligned}
Y_{it} = & \beta_0 + \beta_1 \widehat{PrSub_{lt}} + \beta_2 MaxSub_{it} \\
& + \beta_3 (\widehat{PrSub_{lt} \times MaxSub_{it}}) + \beta_4 X_{it} \\
& + \eta_c + \eta_t + \varepsilon_{it}
\end{aligned} \tag{2}
$$

- Y_{it}: outcome
  - Indicator for a 3+/4+/5-star provider in tract (quality)
  - log out-of-pocket payment at provider (OOP)

Introduction

Context and Data

Quasi-Experimental Evidence
- Quasi-Experimental Methods
- Quality
- Out-of-Pocket Payments

Model

Estimation and Counterfactuals

Conclusion

2SLS-DiD: Subsidy Access Raises Quality

	3+ Stars	4+ Stars	5 Stars
Prop. Fam. Subsidized (1pp)	0.044***	0.018***	0.013**
	(0.009)	(0.007)	(0.005)
Max. Sub. ($10/wk)	-0.050*	-0.001	0.011
	(0.027)	(0.018)	(0.014)
Prop. Fam. Sub. × Max. Sub.	-0.007***	-0.002	-0.001
	(0.002)	(0.001)	(0.001)
N	5,319	5,319	5,319
F (Prop. Fam. Sub.)	36.0	36.0	36.0
F (Prop. Fam. Sub. × Max. Sub.)	34.7	34.7	34.7

Instrument for proportion subsidized and its interaction with the maximum subsidy using distance to income verification office and its interaction with the maximum subsidy. Sample is Wisconsin tracts observed from 2017-2020. Control for local SNAP receipt for families with children under age five, income distribution for families with young children, marital status of parents with young children, birth rates by marital/education status, urbanicity, number of young children in the tract. Results weighted by children under age five. Cluster standard errors by county-year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Introduction

Context and Data

Quasi-Experimental Evidence
- Quasi-Experimental Methods
- Quality
- Out-of-Pocket Payments

Model

Estimation and Counterfactuals

Conclusion

2SLS-DiD: Max. Subsidy Increases Out-of-Pocket Payments

	ln(OOP)	OOP>0	ln(Price)
Prop. Fam. Subsidized (1pp)	0.013	0.006	0.002*
	(0.012)	(0.004)	(0.001)
Max. Sub. ($10/wk)	0.181**	0.066**	0.051***
	(0.085)	(0.026)	(0.009)
Prop. Fam. Sub. $\times$ Max. Sub.	-0.003	-0.001*	-0.000
	(0.002)	(0.001)	(0.000)
N	7,557	7,557	7,557
F (Prop. Fam. Sub.)	34.4	34.4	34.4
F (Prop. Fam. Sub. $\times$ Max. Sub.)	42.0	42.0	42.0

Instrument for proportion subsidized and its interaction with the maximum subsidy using distance to income verification office and its interaction with the maximum subsidy. Sample is Wisconsin licensed private providers, from 2017-2020. Outcome is (1) log out-of-pocket payment plus one; (2) whether the out-of-pocket payment is greater than zero; (3) log sticker price. Results are provider-level and weighted by subsidized enrollment. Control for local SNAP receipt for families with children under age five, income distribution for families with young children, marital status of parents with young children, birth rates by marital/education status, urbanicity, number of young children in the tract, quality rating, county, provider-type FE. Cluster standard errors by county-year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Takeaways

- Tracts with more subsidized families are more likely to have high-quality care

Takeaways

- Tracts with more subsidized families are more likely to have high-quality care
- Providers with higher maximum subsidies have higher OOPs, prices

Takeaways

- Tracts with more subsidized families are more likely to have high-quality care
- Providers with higher maximum subsidies have higher OOPs, prices
- Similar patterns using alternative approaches Triple-Diff PA/WI DiD PA/WI Quality PA/WI OOPs Price Histogram

Takeaways

- Tracts with more subsidized families are more likely to have high-quality care
- Providers with higher maximum subsidies have higher OOPs, prices
- Similar patterns using alternative approaches Triple-Diff PA/WI DiD PA/WI Quality PA/WI OOPs Price Histogram
- We still don't know:
  - How many fewer families are subsidized if the maximum subsidy increases?
  - Are price and quality changes as important as family choices?

Introduction

Context and Data

Quasi-Experimental Evidence

Model

Estimation and Counterfactuals

Conclusion

Overview

- Market equilibrium determined through three-stage game:
  1. Providers choose their quality rating

Overview

- Market equilibrium determined through three-stage game:
  1. Providers choose their quality rating
  2. Providers observe market quality and set prices

Overview

- Market equilibrium determined through three-stage game:
  1. Providers choose their quality rating
  2. Providers observe market quality and set prices
  3. Families choose provider in the market, differentiated by characteristics

Overview

- Market equilibrium determined through three-stage game:
  1. Providers choose their quality rating
  2. Providers observe market quality and set prices
  3. Families choose provider in the market, differentiated by characteristics

- State serves as many families as it can afford, taking maximum subsidy and budget as exogenous

Introduction

Context and Data

Quasi-Experimental Evidence

Model
- Demand
- Supply
- Balanced Budget
- Implications

Estimation and Counterfactuals

Conclusion

Demand

- Families choose a provider to maximize utility Utility

Demand

- Families choose a provider to maximize utility Utility
- Choice set varies across families
  - Market-year
  - Child's age
  - Income-based eligibility (Head Start)

Demand

- Families choose a provider to maximize utility Utility
- Choice set varies across families
- Utility depends on characteristics common to all families
  - Provider type
  - Quality rating
  - Years in operation
  - Other characteristics
  - Unobserved average demand shock

Demand

- Families choose a provider to maximize utility Utility
- Choice set varies across families
- Utility depends on characteristics common to all families
- Utility depends on characteristics and preferences varying by family
  - Distance
  - Out-of-pocket payment (depends on subsidy status)
  - Preferences for provider characteristics vary by subsidy eligibility
  - Unobserved taste for quality

Introduction

Context and Data

Quasi-Experimental Evidence

Model
- Demand
- Supply
- Balanced Budget
- Implications

Estimation and Counterfactuals

Conclusion

Providers Choose Quality to Maximize Expected Profit

- First, provider expects to earn different profit depending on its quality rating, given what it knows about itself and the market
  - Own characteristics
  - Other providers' characteristics
  - The shape of demand
  - The distribution of private fixed costs
  - Own fixed cost

Providers Choose Quality to Maximize Expected Profit

- First, provider expects to earn different profit depending on its quality rating, given what it knows about itself and the market
- For illustration, consider two quality ratings: *Hi* and *Lo* 5 Types Case Max Problem
  - Draws fixed cost of different quality ratings
  - Average fixed cost of *Lo* (c_{Lo}) is normalized to zero

Providers Choose Quality to Maximize Expected Profit

- First, provider expects to earn different profit depending on its quality rating, given what it knows about itself and the market
- For illustration, consider two quality ratings: *Hi* and *Lo* 5 Types Case Max Problem
- Probability of being high quality:

$$Pr(Qual_{jt} = Hi) = \frac{\exp(\mathbb{E}\Pi_{jt}(Hi) - c_{Hi})}{\exp(\mathbb{E}\Pi_{jt}(Hi) - c_{Hi}) + \exp(\mathbb{E}\Pi_{jt}(Lo))}$$

Providers Choose Quality to Maximize Expected Profit

- First, provider expects to earn different profit depending on its quality rating, given what it knows about itself and the market
- For illustration, consider two quality ratings: *Hi* and *Lo* 5 Types Case Max Problem
- Probability of being high quality:

$$Pr(Qual_{jt} = Hi) = \frac{\exp(\mathbb{E}\Pi_{jt}(Hi) - c_{Hi})}{\exp(\mathbb{E}\Pi_{jt}(Hi) - c_{Hi}) + \exp(\mathbb{E}\Pi_{jt}(Lo))}$$

- More likely to be *Hi* when $\mathbb{E}\Pi_{jt}(Hi)$ rises relative to $\mathbb{E}\Pi_{jt}(Lo)$, or when fixed cost c_{Hi} falls

Providers Set Prices to Maximize Profits

- Second, given quality and anticipating demand, providers set prices to maximize profits

Providers Set Prices to Maximize Profits

- Second, given quality and anticipating demand, providers set prices to maximize profits
- From FOCs:

Maximization Problem

$$p_{jt} = mc_{jt} + \underbrace{\left| \frac{sh_{jt}}{\frac{\partial sh_{jt}}{\partial p_{jt}}} \right|}_{\text{mark-up}}$$

Providers Set Prices to Maximize Profits

- Second, given quality and anticipating demand, providers set prices to maximize profits
- From FOCs:

Maximization Problem

$$p_{jt} = mc_{jt} + \underbrace{\left| \frac{sh_{jt}}{\frac{\partial sh_{jt}}{\partial p_{jt}}} \right|}_{\text{mark-up}}$$

- Mark-up depends on market share/enrollment (sh_{jt}) and price sensitivity ($\frac{\partial sh_{jt}}{\partial p_{jt}}$)

Introduction

Context and Data

Quasi-Experimental Evidence

Model
- Demand
- Supply
- Balanced Budget
- Implications

Estimation and Counterfactuals

Conclusion

Balanced Budget

- Total budget and maximum subsidy are both set exogenously
- Number of subsidies depends on per-subsidy payment Equation

Balanced Budget

- Total budget and maximum subsidy are both set exogenously
- Number of subsidies depends on per-subsidy payment Equation
  - Operates like a waitlist: offer subsidies until funds run out

Balanced Budget

- Total budget and maximum subsidy are both set exogenously
- Number of subsidies depends on per-subsidy payment Equation
  - Operates like a waitlist: offer subsidies until funds run out
  - Per-subsidy payment increases if:
    - Maximum subsidy increases (for providers priced above old max)

Balanced Budget

- Total budget and maximum subsidy are both set exogenously
- Number of subsidies depends on per-subsidy payment Equation
  - Operates like a waitlist: offer subsidies until funds run out
  - Per-subsidy payment increases if:
    - Maximum subsidy increases (for providers priced above old max)
    - Holding prices fixed, families choose more expensive providers (subsidized families pay nothing up to the maximum subsidy)

Balanced Budget

- Total budget and maximum subsidy are both set exogenously
- Number of subsidies depends on per-subsidy payment Equation
  - Operates like a waitlist: offer subsidies until funds run out
  - Per-subsidy payment increases if:
    - Maximum subsidy increases (for providers priced above old max)
    - Holding prices fixed, families choose more expensive providers (subsidized families pay nothing up to the maximum subsidy)
    - Holding enrollment fixed, prices increase (providers respond to higher demand)

Introduction

Context and Data

Quasi-Experimental Evidence

Model
- Demand
- Supply
- Balanced Budget
- Implications

Estimation and Counterfactuals

Conclusion

Subsidies Affect Family Choices

- A subsidy lowers out-of-pocket payments, narrowing price differences across providers OOP Distribution OOP Change
  - Families choosing providers priced below the maximum pay nothing
  - Out-of-pocket payment falls most for providers priced above the maximum

Subsidies Affect Family Choices

- A subsidy lowers out-of-pocket payments, narrowing price differences across providers OOP Distribution OOP Change
- Subsidized families differentiate between providers by their quality and other non-price characteristics

Subsidies Affect Family Choices

- A subsidy lowers out-of-pocket payments, narrowing price differences across providers OOP Distribution OOP Change
- Subsidized families differentiate between providers by their quality and other non-price characteristics
- The extent to which subsidizing fewer families reduces quality of enrollment depends on preferences and available options

Subsidies Affect Market Offerings

- Reducing the number of subsidized families may lower prices
  - Demand falls for formal care and providers with desirable characteristics
  - Subsidized families are price-insensitive below the maximum subsidy

Subsidies Affect Market Offerings

- Reducing the number of subsidized families may lower prices
- Providers are less likely to be high-quality if the returns to quality fall

Subsidies Affect Market Offerings

- Reducing the number of subsidized families may lower prices
- Providers are less likely to be high-quality if the returns to quality fall
- Maximum subsidy has the opposite effect: which wins out depends on parameters

Introduction

Context and Data

Quasi-Experimental Evidence

Model

Estimation and Counterfactuals
- Estimation
- Counterfactuals

Conclusion

Estimation Overview

- Estimate demand, MC using nested GMM to match aggregate shares and micro-moments (Berry, Levinsohn, and Pakes 1995; Conlon and Gortmaker 2020; Petrin 2002)

BLP algorithm | Nodes | Instruments | Micro-moments | MC spec

- Estimate quality-stage parameters by nested pseudo-likelihood, matching quality distribution consistent with estimated returns (Aguirregabiria and Mira 2007; Sweeting 2013; Bodéré 2023)

Detail

Model Estimates

- Low-income, Milwaukee families are more sensitive to distance
- Low-income families are more sensitive to out-of-pocket payments
- Conditional on rating, families prefer higher-quality providers, but significant heterogeneity Demand parameters

Model Estimates

- Low-income, Milwaukee families are more sensitive to distance
- Low-income families are more sensitive to out-of-pocket payments
- Conditional on rating, families prefer higher-quality providers, but significant heterogeneity Demand parameters
- Marginal costs (roughly) rise with quality MC parameters
- Centers pay lower average fixed costs for higher ratings than family providers Quality-stage parameters Fit (private) Fit (public 4-K)

Introduction

Context and Data

Quasi-Experimental Evidence

Model

Estimation and Counterfactuals
- Estimation
- Counterfactuals

Conclusion

Overview: Effect of Maximum Subsidy Increase

- What was the effect of Wisconsin's 2022 max subsidy increase?
  - In response to a federal budget increase
  - Relative to using same budget to subsidize more families Coverage

Overview: Effect of Maximum Subsidy Increase

- What was the effect of Wisconsin's 2022 max subsidy increase?
  - In response to a federal budget increase
  - Relative to using same budget to subsidize more families Coverage
- Decompose effects into three channels
  - Family choices (holding prices and quality fixed)

Overview: Effect of Maximum Subsidy Increase

- What was the effect of Wisconsin's 2022 max subsidy increase?
  - In response to a federal budget increase
  - Relative to using same budget to subsidize more families Coverage
- Decompose effects into three channels
  - Family choices (holding prices and quality fixed)
  - Prices (holding quality fixed)

Overview: Effect of Maximum Subsidy Increase

- What was the effect of Wisconsin's 2022 max subsidy increase?
  - In response to a federal budget increase
  - Relative to using same budget to subsidize more families Coverage
- Decompose effects into three channels
  - Family choices (holding prices and quality fixed)
  - Prices (holding quality fixed)
  - Quality

Larger Subsidy Reduces Formal Care Enrollment

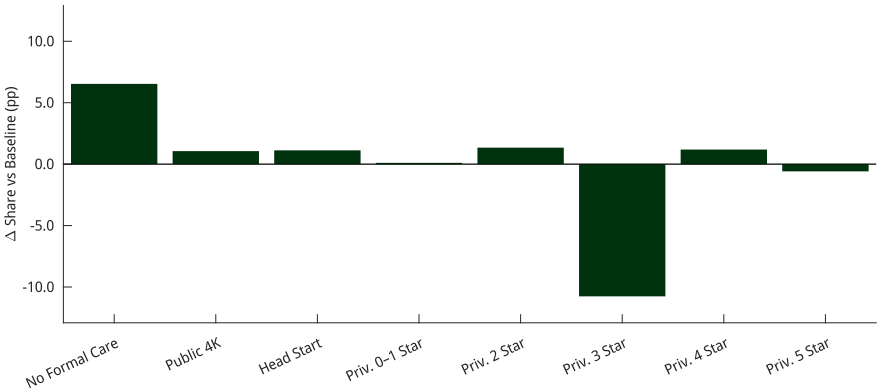


Figure: The Effect of Higher Max on Enrollment Relative to Subsidizing More Families

Prices Explain ∼3/5 of Lower Formal Care Use

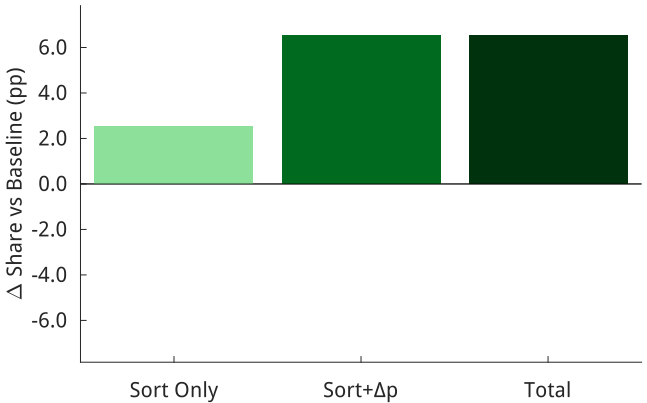


Figure: The Effect of Higher Max on Forgoing Formal Care Relative to Subsidizing More Families

Public Enrollment Rises Mostly Due to Reduced Crowd-Out

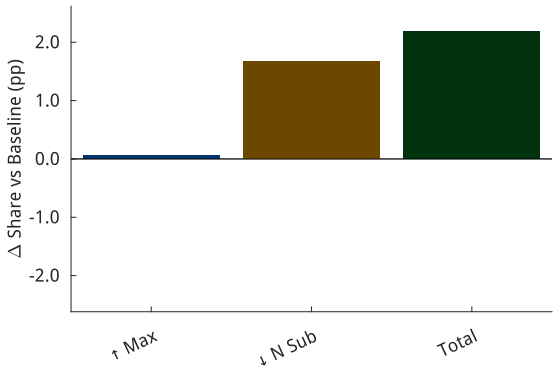


Figure: The Effect of Higher Max on Public Enrollment Relative to Subsidizing More Families

Middling-Quality Enrollment is Lower Under Higher Max

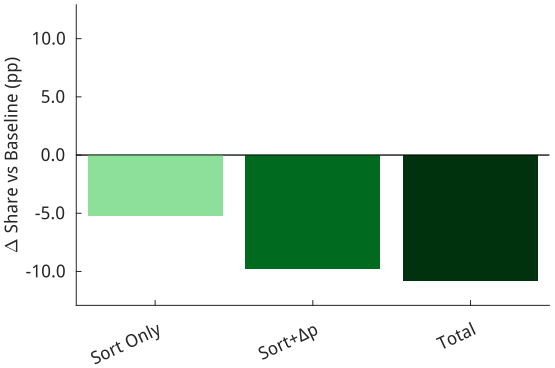


Figure: The Effect of Higher Max on Middling-Quality Enrollment Relative to Subsidizing More Families

2 Star | 4 Star | 5 Star | Quality Distribution | Subsidized Families

Consumer Surplus Falls Under the Larger Subsidy

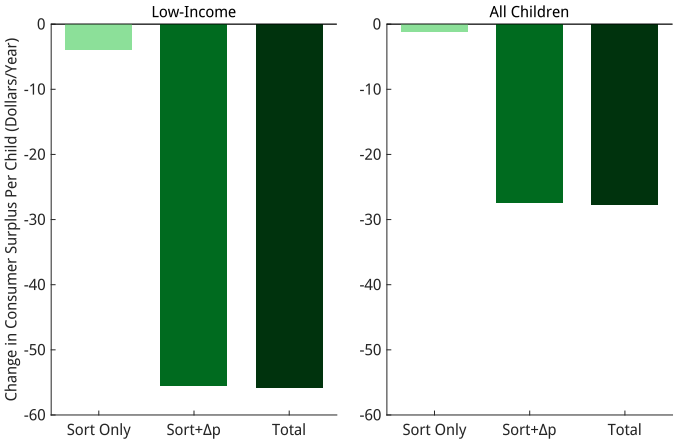


Figure: The Effect of Higher Max on Consumer Surplus Relative to Subsidizing More Families

Calculation

Producer Surplus Also Falls Under the Larger Subsidy

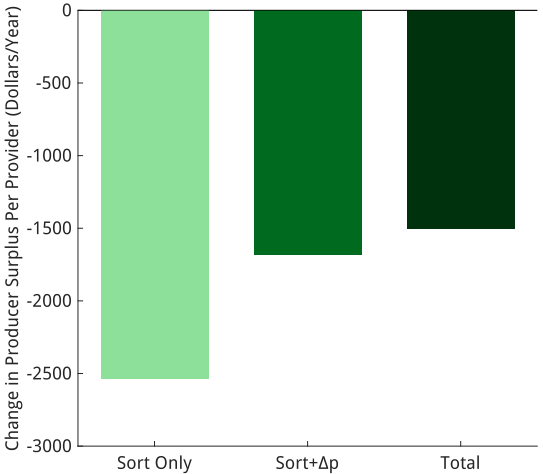


Figure: The Effect of Higher Max on Producer Surplus Relative to Subsidizing More Families

Introduction

Context and Data

Quasi-Experimental Evidence

Model

Estimation and Counterfactuals

Conclusion

Conclusion

- I showed that a higher maximum subsidy pushes low-income children out of formal care by increasing prices

Conclusion

- I showed that a higher maximum subsidy pushes low-income children out of formal care by increasing prices
- Program participation across space is a relevant channel — implications for subsidies in other industries

Conclusion

- I showed that a higher maximum subsidy pushes low-income children out of formal care by increasing prices
- Program participation across space is a relevant channel — implications for subsidies in other industries
- Part of ongoing work:
  - Effect of changing the fixed costs of program enrollment (e.g., ordeals and targeting)
  - Entry, exit, and capacity constraints
  - Endogenous costs (e.g., child care worker labor market)

Thank you!

serena.goldberg@yale.edu

CCDF Purposes (per CCDBG Act of 1990)

1. **To allow each State maximum flexibility** in developing child care programs and policies that best suit the needs of children and parents within that State;
2. To promote parental choice to empower working parents to make their own decisions [...];
3. To encourage States to provide consumer education information to help parents make informed choices [...];
4. To assist States in delivering high-quality, coordinated early childhood care and education services to maximize parents' options and support parents trying to achieve independence from public assistance;
5. **To assist States in improving the overall quality of child care services and programs** [...];
6. To improve child care and development of participating children; and
7. **To increase the number and percentage of low-income children in high-quality child care settings.**

Back

LATE Interpretation of Effect of Proportion Subsidized

- LATE, weighting tracts most responsive to distance instrument
  (Angrist, Graddy & Imbens 2000)
- Requires:
  - Relevant
  - Exogenous
  - Monotonicity

Back

ATE Interpretation of Effect of Max Subsidy

- Under constant effects, average treatment effect
- Requires parallel trends, no anticipation
- Under heterogeneous effects: TWFE weights can be negative (de Chaisemartin & D'Haultfœuille 2020)

Back

Demand Parameter Estimates

Parameter	Coef.	SE	Parameter	Coef.	SE
2 Stars	−6.2114	2.5969	Eligible	−3.4020	0.2327
3 Stars	−10.9830	2.5846	2 Stars × Elig	2.1306	0.1822
4 Stars	−2.4627	3.0677	3 Stars × Elig	10.9624	0.2687
5 Stars	3.4796	2.8199	4 Stars × Elig	−0.5501	0.1874
Center-Based	−0.5531	1.1664	5 Stars × Elig	−7.6373	0.2770
Capacity	0.1022	0.0118	Center × Elig	14.9996	0.3372
Youngest Age	0.2263	0.0412	OOP × Elig	−4.5385	0.0974
Oldest Age	−0.1254	0.0191	OOP × Inelig	−2.2864	0.0513
Years in Op.	−0.7484	0.0481	Distance × Elig	−14.9992	0.3695
Inside Option	125.8478	4.7205	Distance × Inelig	−0.2213	0.2091
Head Start	−68.4632	3.2737	Distance^2	−10.1714	0.2370
Public 4K	−35.6361	4.3703	Dist × Milw × Elig	−0.0648	1.1175
Pub 4K FDHW	77.5105	6.3138	Dist × Milw × Inelig	−2.4916	0.6755
Pub 4K HDFW	5.0965	2.9320	Elig × Public 4K	−10.4724	0.4129
Pub 4K HD4D	8.7906	3.2249	Elig × Pub 4K FDHW	−2.7503	2.0321
Public Fee-Based	−26.0161	2.6787	Elig × Pub 4K HDFW	−6.3655	0.5355
Age 0–2 (inside)	8.3895	0.2930	Elig × Pub 4K HD4D	−6.5367	0.4403
Age 3 (inside)	18.2254	0.5050	Elig × Pub 4K Fee	6.6290	0.2354
Age 4 (inside)	0.7797	0.1912	σ (High-Quality RC)	9.3260	0.2312

Back

Marginal Cost Parameter Estimates

Parameter	Coef.	SE	Parameter	Coef.	SE
2 Stars	0.3910	0.1754	Oldest Age Served	−0.0022	0.0113
3 Stars	0.2588	0.1840	Oldest Age^2	0.0000	0.0000
4 Stars	0.4551	0.2261	Years in Op.	−0.0214	0.0114
5 Stars	1.9998	0.1741	Years in Op.^{2}	0.0011	0.0003
Local Wages	−0.0093	0.0026	Center-Based	−0.7606	0.1038
Youngest Age	0.0347	0.0152	exp(Utilization)	0.1880	0.0643
Youngest Age^2	−0.0009	0.0004			

Back

Quality-Stage Parameter Estimates

	Family Non-Milw.	Family Milwaukee	Center Non-Milw.	Center Milwaukee
θ^{ε}	14.97	–	–	–
c_2	0.00	0.00	0.00	0.00
c_3	4.05	0.00	0.00	0.00
c_4	20.86	29.13	23.65	18.66
c_5	23.88	38.78	5.32	7.34

θ^{ε} is the scale parameter for the distribution of private fixed costs. c_k is the additional cost of being a k star provider compared to being unrated. θ^{ε} does not vary by provider type or whether the provider is in Milwaukee. I assume that fixed costs to be quality-rated are at least zero, but otherwise am agnostic on the distribution of fixed costs.

Back

Model Fit: Private Enrollment

	Subsidized		Unsubsidized		
Moment	Data	Model	Data	Model	Matched
Prop. 0/1 Stars	0.00	0.00	0.06	0.00	
Prop. 2 Stars	0.38	0.28	0.25	0.28	✓
Prop. 3 Stars	0.36	0.53	0.35	0.39	✓
Prop. 4 Stars	0.06	0.05	0.07	0.05	✓
Prop. 5 Stars	0.20	0.12	0.28	0.20	✓
Prop. Center-based	0.82	0.85	0.90	0.74	
Avg. Provider Age	12	17	17	17	
Avg. Weekly Sticker Price	240	200	210	170	
Avg. Weekly OOP	18	7.8	210	170	✓

Statistics over families choosing a private provider. OOP is weekly out-of-pocket, net of subsidy for subsidized families. Checkmarks indicate GMM-targeted moments.

Back

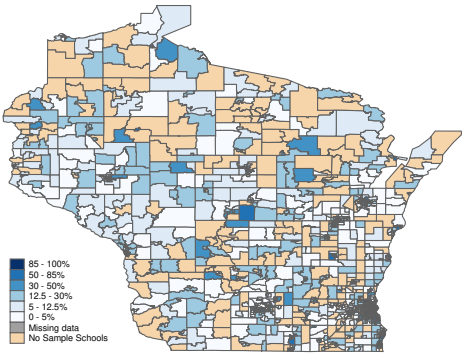
Model Fit: Share FRPL-Eligible in Public 4-K

Public 4-K Subset	Data	Model	Matched
All Public 4-K	0.38	0.54	✓
Charter	0.51	0.52	
Full Day, Full Week	0.54	0.59	
Full Day, Half Week	0.42	0.57	
Half Day, Full Week	0.31	0.49	
Half Day, 4 Days/Week	0.36	0.52	

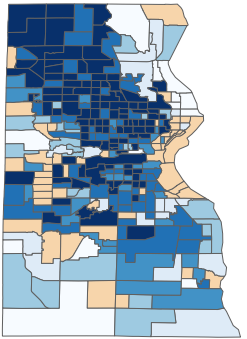
Share of children enrolled in each type of Public 4-K program eligible for Free/Reduced Price Lunch.

Back

Subsidy Use is Concentrated Across Space



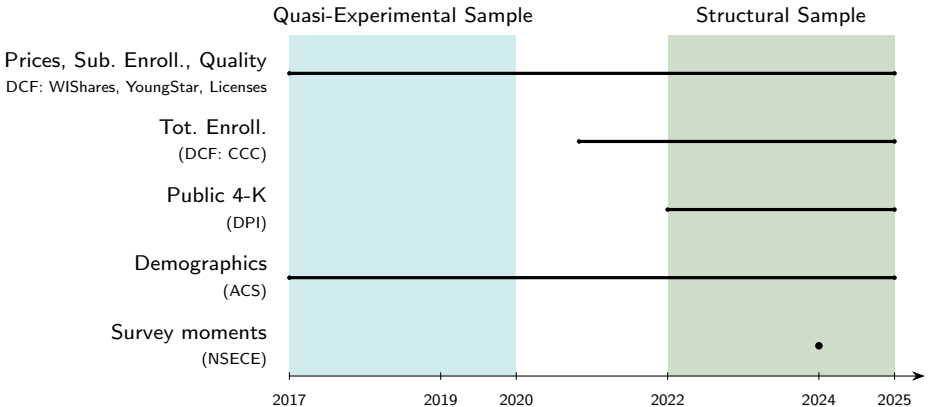
(a) Wisconsin



(b) Milwaukee

Figure: Share of Private Enrollees with Subsidies, by Tract (2022-2025) Back

Data Availability Timeline



Back (QE) Back (Structural)

Price Data

- WI Dept. of Children and Families
- Starting in 4/2017, switch to EBT cards
- Prices collected when provider agrees to accept subsidies, can be updated at any time
- May vary by age

Back

Subsidy Enrollment Data

- WI Dept. of Children and Families
- Starting in 4/2017, switch to EBT cards to manage subsidy program
- Subsidized enrollment observed monthly for every provider

Back

Quality Rating Data

- Discrete 1–5 star rating, updated annually
- 2 stars: meet licensing + accept subsidies
- 3+ stars: min. standards for curriculum, process/structural quality, staff education
- 4-5 stars: average score of 4-5 on Environment Rating Scale
- WI: 87% of licensed private providers quality-rated

Back

Provider Characteristics Data

- WI Dept. of Children and Families provider licenses
- Address, licensed hours/months, ages served, license start/end, capacity, facility type
- Facility type:
  - Commercial = *centers*
  - Residential = *family providers*
  - Public school = *fee-based publics*
- Licensed capacity = upper bound (does not reflect staffing/ratio)

Total Enrollment Data

- WI Child Care Counts program
- Grants distributed based on monthly enrollment + employment reports
- Enrollment available for 86.4% of in-sample seats

Back

Public 4-K Data

- WI Dept. of Public Instruction (DPI)
- Annual September enrollment counts by grade
- Enrollment broken out by Free/Reduced Price Lunch (FRPL) eligibility
- FRPL threshold ($\leq$185% FPL) $\approx$ subsidy eligibility
- Also: addresses, schedules (32% full-day/full-week; 58% half-day), charter status, whether offer K3

Back

Survey Moments (NSECE)

- National Survey of Early Care and Education (2024)
- Match moments on distance, age to provider by income (e.g., would be eligible for subsidies)
- Used only for the structural model

Back

Structural Sample: Provider Types

	WI Model (2022–2025)	WI Lic.+Pub (2022–2025)	In-Sample+Miss. (2022–2025)
Family provider	0.056	0.056	0.060
Public: Fee-Based	0.030	0.029	0.029
Public 4-K	0.306	0.271	0.304
Head Start	0.061	0.055	0.060

Enrollment-weighted shares of the full sample.

Back

Structural Sample: Private Providers

	WI Model (2022–2025)	WI Lic. (2022–2025)	In-Sample+Miss. (2022–2025)
Quality rating	3.10	2.89	3.03
Sticker price, mo. ($2017)	849.40	815.90	849.40
OOP, subsidized (mo., $2017)	70.83	70.35	71.42
Cap.-constrained (lic. only)	0.348	0.383	0.352
Min. age accepted (months)	5.5	11.5	5.5
Max. age accepted (months)	144.1	143.6	143.8
Serves school-age only (5+)	0.000	0.068	0.000
Years of operation	15.2	15.2	15.0
Total est. sub. enrollment	98,686	120,780	107,420
Total est. enrollment	346,874	423,500	355,737

Enrollment-weighted. Excludes Public 4-K, Head Start, fee-based publics.

Back

Structural Sample: Public 4-K Detail

	WI Model (2022–2025)	In-Sample+Miss. (2022–2025)
Full day, full week	0.324	0.324
Half day, full week	0.287	0.287
Full day, half week	0.100	0.100
Half day, 4 days/week	0.289	0.289
Charter school	0.052	0.066
Offers K3	0.068	0.067
Share FRPL-eligible	0.376	0.377

Public 4-K only. "Full day" = school day, not work day.

Back

Sample Comparison: Stylized Facts vs Structural

	WI Model (2022-25)	WI Licensed (2022-25)	WI Samp. (2017-20)	PA Samp. (2017-20)
Family provider	0.064	0.069	0.102	0.050
Public: Fee-Based	0.082	0.105	0.000	0.000
Head Start	0.112	0.109	0.000	0.000
High-quality	0.315	0.259	0.280	0.335
Sticker price, mo. ($2017)	760.15	718.00	909.37	823.55
Avg sticker price if > 0	856.45	811.92	909.37	823.55
No subsidized enrollees	0.135	0.225	0.180	0.023
Serves school-age only (5+)	0.000	0.068	0.000	0.000
Total licensed seats	524,150	715,882	494,491	1,272,832
N providers	10,672	17,874	13,025	22,077

Capacity-weighted. Excludes Public 4-K.

Back

2SLS-DiD First Stage: Interaction

- IV for interaction of proportion subsidized with maximum subsidy

$$
\begin{aligned}
(PrSub_{lt} \times MaxSub_{it}) = & \gamma_0 + \gamma_1 DistOffice_i + \gamma_2 MaxSub_{it} \\
& + \gamma_3 (DistOffice_i \times MaxSub_{it}) + \gamma_4 X_{it} \quad (1b) \\
& + \zeta_c + \zeta_t + \nu_{it}
\end{aligned}
$$

- Instrument: $DistOffice_i \times MaxSub_{it}$

Back

Triple-Diff Quality

	3+ Stars	4+ Stars	5 Stars
$\Delta \ln PrSub \times$ Post	0.099***	0.025	0.014
	(0.030)	(0.015)	(0.013)
Max. Sub. $\times$ Post	0.004**	0.005***	0.002
	(0.002)	(0.002)	(0.003)
$\Delta \ln PrSub \times$ Max. Sub. $\times$ Post	0.007	-0.005	0.003
	(0.008)	(0.004)	(0.005)
N	5,324	5,324	5,324

$\Delta \ln PrSub_i$ is log change in proportion subsidized between 2017–18 and 2019–20. Max. Sub. is 2019–2020 average maximum subsidy in \$10/wk (mean-centered). Only interactions with Post shown. Sample is Wisconsin tracts observed from 2017–2020. Control for local SNAP receipt for families with children under age five, income distribution for families with young children, marital status of parents with young children, birth rates by marital/education status, urbanicity, number of young children in the tract, county FE. Results weighted by children under age five. Cluster standard errors by county-year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Triple-Diff OOP

	ln(OOP)	OOP>0	ln(Price)
$\Delta \ln PrSub \times$ Post	-0.107	-0.013	-0.001
	(0.130)	(0.045)	(0.008)
Max. Sub. $\times$ Post	0.038***	0.011***	0.001**
	(0.012)	(0.004)	(0.001)
$\Delta \ln PrSub \times$ Max. Sub. $\times$ Post	-0.029	-0.015**	-0.002
	(0.020)	(0.007)	(0.002)
N	7,556	7,556	7,556

$\Delta \ln PrSub_i$ is log change in proportion subsidized between 2017–18 and 2019–20. Max. Sub. is 2019–2020 average maximum subsidy in \$10/wk (mean-centered). Only interactions with Post shown. Sample is Wisconsin licensed private providers, from 2017-2020. Weighted by subsidized enrollment. Cluster SEs by county-year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Alt Design: Compare High- vs. Low-Poverty Tracts, WI vs. PA

$$Y_{it} = \beta_0 + \beta_1^{state} HighPov_i + \beta_2^{state} Post_t + \beta_3^{state} HighPov_i \cdot Post_t + \beta_4 X_{it} + \eta_{county(i)} + \epsilon_{it}$$

- Motivation: in 2019 both WI and PA responded to a federal funding increase by changing their subsidy programs
- β_3^{state}: state-specific policy effect
- $HighPov_i$: top 25% of tracts by poverty share (state-specific)
- X_{it}: demographics, provider controls, pre-period care availability

Alt Design: PA Policy More Effective at Raising Quality

	P(Any HQ Provider)
WI × Post × High Poverty	0.000
	(0.014)
PA × Post × High Poverty	0.022**
	(0.010)
Observations	17093

Standard errors clustered by county-year. Weighted by number of children under age five. Outcome: tract has any 4+ star provider in WI, any 3+ star in PA (both based on same ERS cut-off). Dependent mean in WI high-poverty tracts in 2018: 0.289. In PA: 0.285. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Back

Alt Design: Prices Cluster Near Max. Sub. in High-Poverty Tracts

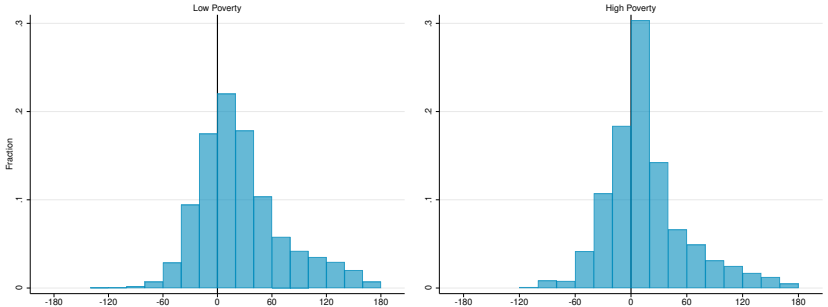
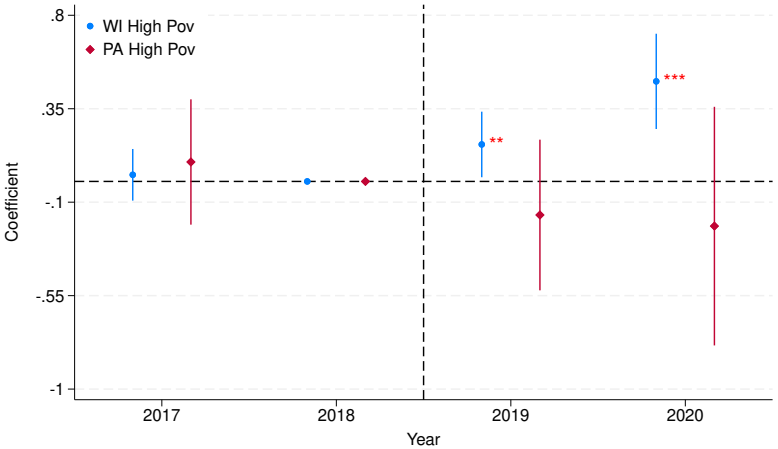


Figure: Prices Relative to Maximum Subsidy Rate, By Census Tract Poverty

Back

Alt Design: OOPs Rise in High-Poverty WI Tracts Post-2019



Family Utility

$$U_{ijt}(sub) = \beta_i^X X_{jt} + \beta_i^Q Qual_{jt} + \beta_i^{oop} OOP_{jt}(sub; MaxSub) + \beta_i^d d_{ijt} + \xi_{jt} + \epsilon_{ijt}$$

- Include characteristics X_{jt}, quality $Qual_{jt}$, and distance d_{ijt}
- OOP: out-of-pocket payment, depends on whether the family is subsidized and the maximum subsidy
- ξ_{jt}: average unobserved demand shock for j, time t
- ϵ_{ijt}: idiosyncratic taste
- Subsidies shift demand through the out-of-pocket payment

Back

Market Share for Provider j

$$sh_{jt} = \sum_{\tau} \sum_{l \in \mathcal{L}(t)} w_{lt}^{\tau} \pi_t^{\tau} \int \frac{\exp(\delta_{jt} + \mu_{ijt}(\tau, MaxSub))}{\sum_k \exp(\delta_{kt} + \mu_{ikt}(\tau, MaxSub))} dF(\nu)$$

- For now, let's say $\tau \in \{\text{subsidized, unsubsidized}\}$
- Shifting families from unsubsidized to subsidized increases π_t^{sub} and decreases π_t^{unsub}
- Subsidies shift demand towards more expensive, desirable providers

Back

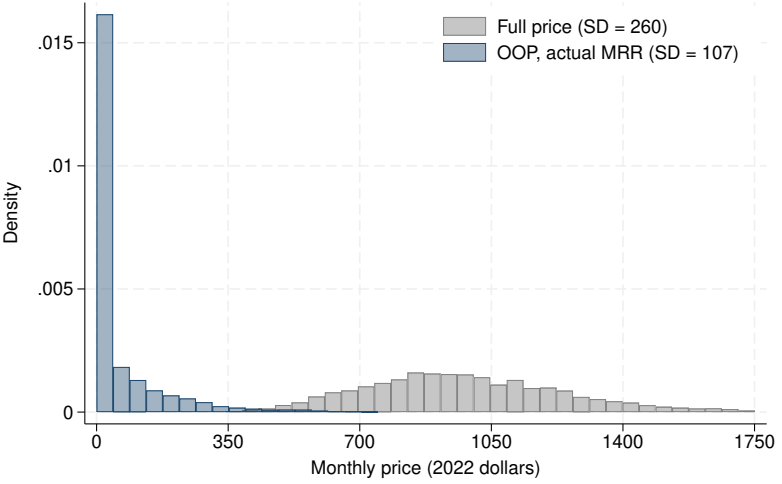
Utility Change: Unsubsidized → Subsidized

$$\Delta_{sub} U_{ijt} = |\beta_i^{oop}| \cdot \min(p_{jt},\ MaxSub)$$

- Newly subsidized family goes from paying p_{jt} to either 0 or $p_{jt} - MaxSub$
- Largest increase for $p_{jt} > MaxSub$ (full subsidy)
- If families are more sensitive to out-of-pocket payments, the change in utility will be larger

Back

OOP Zero-Bound: Prices Compress Below Max. Sub.



Sample: private providers eligible for the subsidy.

OOP Change: *MaxSub* Increase ($MS_0 \rightarrow MS_1$)

$$\Delta_{\rightarrow MS_1} OOP_{jt} = \begin{cases} -(MS_1 - MS_0) & \text{if } p_{jt} > MS_1 \\ -(p_{jt} - MS_0) & \text{if } MS_0 \leq p_{jt} \leq MS_1 \\ 0 & \text{if } p_{jt} < MS_0 \end{cases}$$

Back

$\partial sh_{jt}/\partial p_{jt}$ Decomposition

$$\frac{\partial sh_{jt}}{\partial p_{jt}} = \sum_{\tau} \pi_t^{\tau} \frac{\partial sh_{jt}}{\partial OOP_{jt}^{\tau}} \frac{\partial OOP_{jt}^{\tau}}{\partial p_{jt}}$$

- Say $\tau \in \{sub, unsub\}$
- Subsidized families pay no additional cost as p_{jt} increases if $p_{jt} < MaxSub$ ($\frac{\partial OOP_{jt}^{\tau}}{\partial p_{jt}} = 0$)
- As more families are subsidized, $\pi_t^{sub} \uparrow$, $\pi_t^{unsub} \downarrow \implies \frac{\partial sh_{jt}}{\partial p_{jt}} \downarrow$ for $p_{jt} < MaxSub$

Back

Firm Price-Setting

$$\Pi_{jt}(Qual_{jt}) = \max_{p_{jt}>0} sh_{jt}(p_{jt} - mc_{jt}(Qual_{jt}))$$

$$\implies p_{jt} = mc_{jt} + \left| \frac{sh_{jt}}{\frac{\partial sh_{jt}}{\partial p_{jt}}} \right|$$

- Providers choose price p_{jt} to maximize profit given chosen quality
- Providers account for the effect on enrollment sh_{jt}
- Providers with greater sh_{jt} set a higher mark-up (market power)
- When families are less price-sensitive $\implies$ higher mark-ups

Back

Quality-Choice Maximization Problem

Provider picks q to maximize expected profit net of fixed cost and shock:

$$\max_{q \in \{Lo, Hi\}} \mathbb{E}\Pi(q) - c_q - \varepsilon_{qjt}$$

- $\mathbb{E}\Pi(q)$: expected profit at rating q
- c_q: fixed cost of being rating q (c_{Lo} normalized to 0)
- $\varepsilon_{qjt} \sim$ EV1: idiosyncratic shock, gives the logit CCP on the main slide

Back

Quality Choice: Full 5-Rating Case

Full discrete choice over $q \in \{0, 2, 3, 4, 5\}$ stars:

$$Pr(Qual_{jt} = q) = \frac{\exp(\mathbb{E}\Pi(q) - c_q)}{\sum_{q' \in \mathcal{Q}} \exp(\mathbb{E}\Pi(q') - c_{q'})}$$

- Same logic as the two-rating case: provider more likely to pick higher q when its net return $\mathbb{E}\Pi(q) - c_q$ is higher

Back

Balanced Budget

Total funding Y_t pins down the share of eligible families served:

$$Y_t = Pr(Sub) \sum_{i \in \mathcal{I}(Elig)} \min\{MaxSub, p_{j(i,s=1),t}\}$$

- Y_t: state's total exogenous funding
- $Pr(Sub)$: share of eligible families receiving subsidies
- $p_{j(i,s=1),t}$: price at the provider family i would choose if subsidized
- Works like a waiting list: if $MaxSub$ or provider prices rise, $Pr(Sub)$ falls

BLP Algorithm

1. Given a guess of *i*-specific parameters, find average utility δ_j to match aggregate shares by solving a fixed point problem
2. Given δ_j and *i*-specific parameters, back out unobserved cost shocks from marginal cost FOC
3. GMM on stacked IV moments (ξ_{jt}, ω_{jt}) and micro-moments $\rightarrow$ new guess of *i*-specific parameters
4. Repeat until convergence

Nodes Instruments Micro-moments MC spec

Back

Keeping the Spatial Problem Tractable

- Divide each market into "nodes" at census block centroids
- Distance: space between the node to provider's exact address
- Family locations not directly observed, use ACS data to determine distribution of types over space
- Also add additional micro-moments to identify taste for distance

Back

Instruments

- Exogenous own-characteristics
- BLP instruments: exogenous characteristics of other providers in market
- Spillovers from child care worker wage subsidy program
- Used on supply side only:
  - Expected market share relative to capacity
  - Distance to income-verification offices

Back

Micro-Moments

- By market, subsidy status:
  - Proportion of families in each quality type
  - Average out-of-pocket payment
- By market type (urban/suburban/rural), eligibility status:
  - Share of enrollees traveling 0–1, 1–3, 3–8 miles
  - Share of enrollees who are school-aged
- By market: Share of 4-K enrollees eligible for FRPL

Marginal Cost Specification

$$mc_{jt} = \theta^X X_{jt}^S + \theta^Q Qual_{jt} + \theta^C \cdot \exp(sh_{jt} N_t / C_{jt}) + \omega_{jt}$$

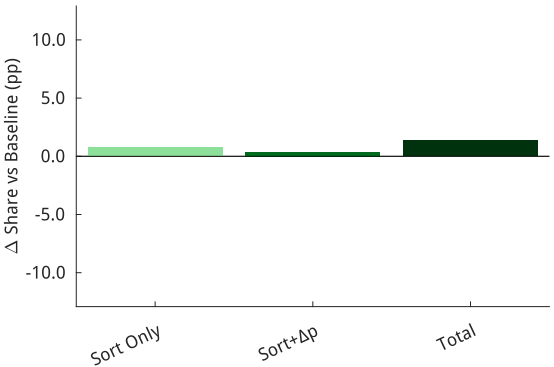
- X_{jt}^S: county wages, min/max age polynomials, provider age polynomial, center indicator
- $Qual_{jt}$: star-rating dummies (2–5)
- $\exp(sh_{jt} N_t / C_{jt})$: increasing in capacity utilization, capturing the shadow cost of the capacity constraint
- ω_{jt}: unobserved MC shock, instrumented Instruments Back

Quality Stage: Nested Pseudo-Likelihood

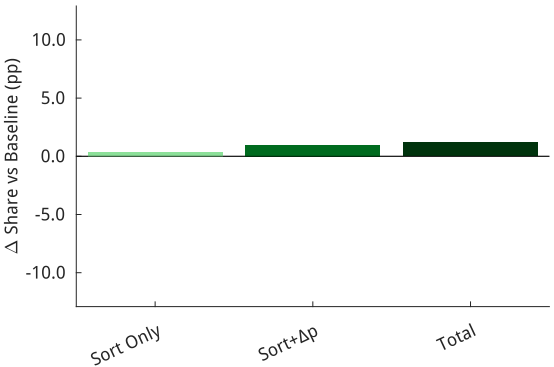
- Choice probabilities depend on $\mathbb{E}\Pi(q)$, fixed cost c_q, scale θ^ε
- Approximate $\mathbb{E}\Pi(q)$ with a neural network over nearby and market-wide provider actions (Bodéré 2023; Sweeting 2013)
- Iterate:
  - Update $\mathbb{E}\Pi(q)$ given current choice probabilities
  - Choose $(c_q, \theta^\varepsilon)$ via MLE
  - Update choice probabilities
- Repeat until convergence

Back

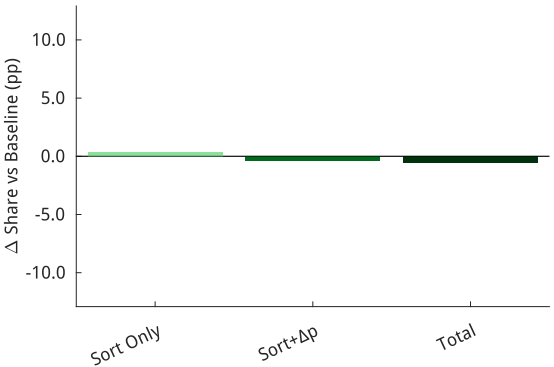
Eligible Families: Enrollment at 2-Star Providers



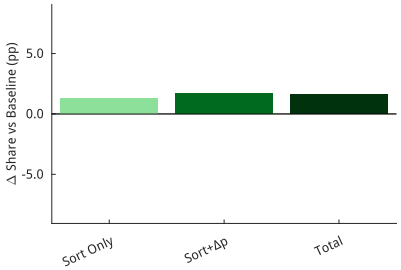
Eligible Families: Enrollment at 4-Star Providers



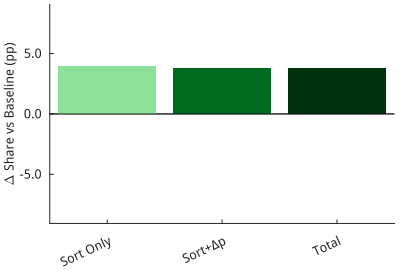
Eligible Families: Enrollment at 5-Star Providers



Subsidized Families Move to the Highest-Quality Providers



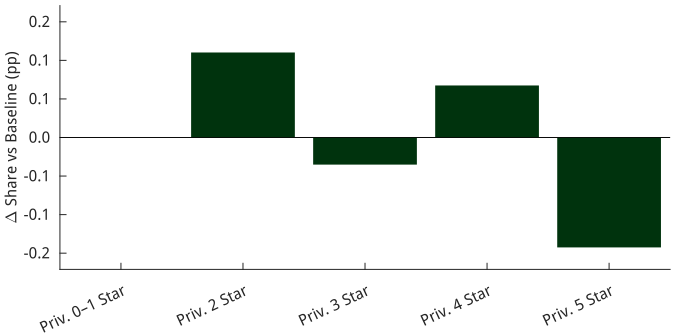
(a) Private 4 Star



(b) Private 5 Star

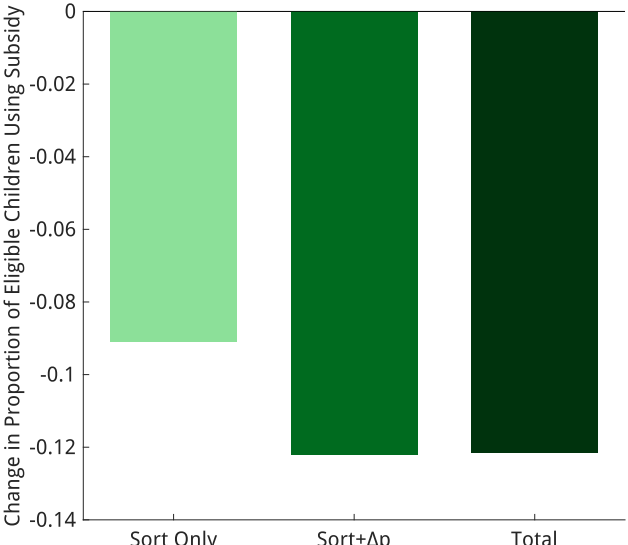
Back

Quality Distribution Shifts Modestly

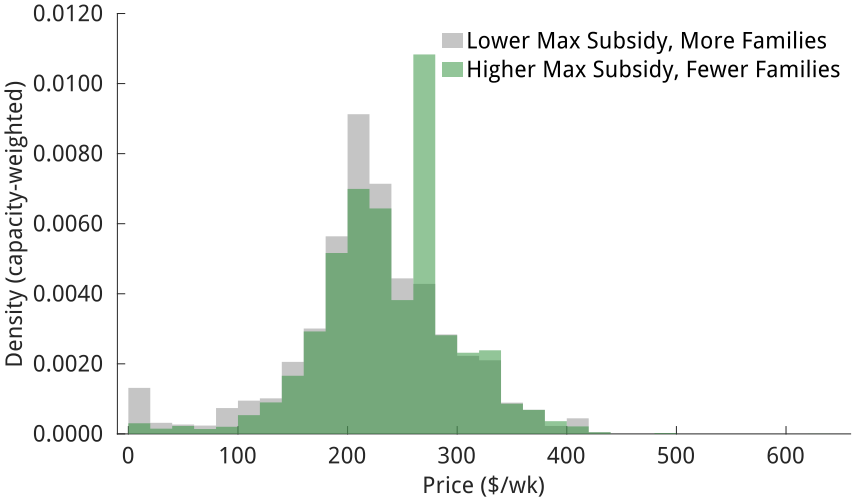


Back

Subsidy Coverage Falls Under the Larger Subsidy



Prices Increase with Maximum Subsidy



Back

CS and PS: Definitions

- Consumer surplus for family i in market t

$$CS_{it} = \frac{1}{|\beta_\tau^{OOP}|} \log \sum_{j \in \mathcal{J}_t} \exp(V_{ijt})$$

  - V_{ijt}: indirect utility of provider j
  - $|\beta_\tau^{OOP}|$ converts utility to dollars

- Producer surplus for private provider j

$$PS_{jt} = sh_{jt} \cdot N_t \cdot (p_{jt} - mc_{jt})$$

  - $sh_{jt} N_t$: total enrollment
  - $p_{jt} - mc_{jt}$: price-cost margin

CS PS

Out-of-Pocket Payments and State Payments

- Family pays out-of-pocket:

$$OOP(p; MaxSub) = \begin{cases} p - MaxSub & \text{if } p > MaxSub \\ 0 & \text{if } p \leq MaxSub \end{cases}$$

- State pays the provider:

$$StatePay(p; MaxSub) = \begin{cases} MaxSub & \text{if } p > MaxSub \\ p & \text{if } p \leq MaxSub \end{cases}$$

Back